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## PAPER

## Temporal interference current stimulation in peripheral nerves is not driven by envelope extraction

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28 April 2023**Abstract**

**Background.** Electrical neuromodulation remains an effective therapy for multiple neurological disorders. One strategy to electrically stimulate nerves utilizes the interference of multiple high frequency waveforms. This technique, known as temporal interference stimulation or interferential current stimulation, has recently gained significant attention as a method to improve the state-of-the-art in neurostimulation in both animal studies and human clinical trials. **Objective.** Here we report our investigation into the fundamental properties of the neuronal response to these types of waveforms—the effects of carrier and envelope frequencies, thresholds, firing behavior, and phase and asymmetric interference patterns. **Methods.** We utilized a cuff electrode on the rat sciatic nerve to apply a variety of interferential signals. We recorded muscle activity in the plantar muscles and biceps femoris, which are proxies for activity on two of the major branches of the sciatic, which are spatially distinct in the target volume. We tested both fundamental recruitment properties as well as spatial techniques to selectively activate either muscle group. **Results.** Our data suggest, contrary to the currently accepted explanation, that neurons do not extract envelopes at all, and that the response to these signals is well explained by a resistor–capacitor (i.e. integrator) membrane with a fixed firing threshold. Basic interference techniques do not change recruitment far from electrodes. Techniques can produce regions of both phasic activation and tonic activation/conduction block. **Conclusions.** An integrator model suggests that interference techniques are less capable of minimally invasive stimulation for a subcortical brain target than previously thought. Human clinical trials using these techniques should reevaluate their methods. Interference stimulation allows significant target selectivity in a peripheral cuff electrode with targets near electrodes. These techniques can allow spatially distinct regions of phasic firing, tonic firing, conduction block, and no effect.

**1. Introduction**

Electrical neuromodulation is the most effective therapy for some neurological disorders including Parkinson's disease and drug-resistant epilepsy [1, 2]. Noninvasive brain stimulation of a deep target, like the hippocampus, has recently been proposed using electrical stimulation based on the interference of high frequency sinusoids. The technique has gone by many names since its inception in 1948, including

interferential current stimulation (ICS), interferential current therapy, and temporal interference stimulation (TIS). The technique is currently the subject of significant research interest, including multiple ongoing human clinical trials [3–14].

Many studies have shown that neurons will respond to signals of lower frequencies (i.e. 200 Hz (phasic), 2000 Hz (tonic)) at lower amplitudes than they will of signals of higher frequencies (i.e. 400 Hz (phasic), 4000 Hz (tonic)), based on well-established

chronaxie/rheobase thresholds [15–18]. Since its inception, interferential techniques have assumed that neurons, when exposed to a signal like the sum 2000 Hz and 2020 Hz, are able to ‘demodulate’ [9], ‘rectify,’ or otherwise ‘extract’ [10] the implied 20 Hz signal. It has been assumed that ‘a resultant current of low frequency is generated’ by this technique [8]. Other studies have focused on plotting and simulating the envelope amplitude [9, 11, 19], assuming that this amplitude is most relevant to predicting firing behaviors. It has been proposed that such nonlinear properties are ‘required’ [10] in order to observe the response to ICS signals. The envelope signal is *not real* (there is no energy at that frequency) but can be ‘extracted’ by nonlinear operators like rectifiers, peak tracing, modulation with an additional signal, or the Hilbert transform, which can generate energy at the lower frequency. Theorizing that such properties are significantly present in neurons, interferential brain stimulation techniques claim to generate phasic stimulation which is stronger in the center of the brain than it is near the electrodes. If low frequency energy is generated in neurons prior to action potential generation, then interference techniques would allow stimulation which is ‘greater than the sum of its parts,’ because the generated low frequency energy has a lower threshold than the higher frequency ‘source’ signals. Our experiments suggest that low frequency energy is either not generated or is not generated in significant enough quantities to dramatically change the neuronal response threshold. Rather, interferential techniques appear to be ‘exactly’ the sum of their parts. Further, our data suggest that the integration and thresholding properties of neuronal membranes are mostly linear processes which do not extract low frequency energy, and any nonlinear properties affect the firing threshold by approximately less than 6%. In addition to our exploration of the phasic on-target firing properties, our data support the conclusions of multiple modeling studies which predicted that interferential techniques will induce significant off-target conduction block due to the source signals [3, 10, 12, 20, 21].

In this work we stimulated the rat sciatic nerve and sought to explore the neuronal activation threshold in response to a variety of parameters: carrier and envelope frequencies, undermodulation ratios, asymmetric and phase interferences, and more. All data observed suggest that neurons do not extract envelopes (or rectify, or demodulate) in significant amounts, and the majority of the neuronal response to interferential signals is described by a linear passive membrane integrator with a fixed firing threshold. This conclusion changes several predictions of the responses to these signals, particularly that amplitude steering works in the opposite direction. Some interference techniques may still allow significant improvements to neuromodulation therapies. Our data suggest that it is possible to

produce multiple stimulation regions inside of a target with different activation frequencies. For example, we show data of activation at the beat frequency, double the beat frequency, or the carrier (i.e. conduction block). We believe this technique will also bear great significance for peripheral nerve stimulators, which are useful in a variety of applications [22–29].

## 2. Methods

### 2.1. 12-Contact peripheral cuff electrode

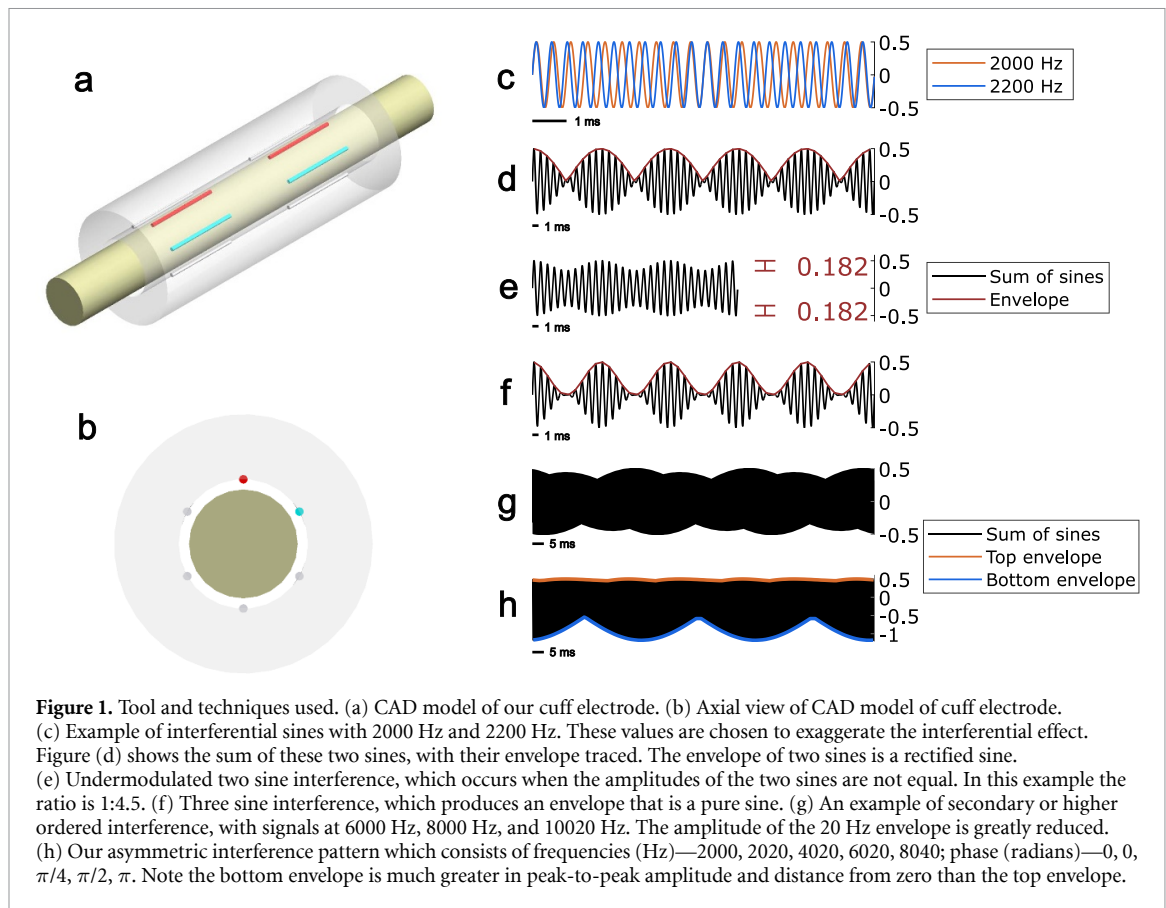
We utilized drawn filled tube wire with a 90% platinum 10% iridium alloy outer sheath (20% of total mass) with a tantalum core (80% of total mass), with a wire diameter of 50  $\mu\text{m}$  and PFA insulation to a total outer diameter of 75  $\mu\text{m}$  (Fort Wayne Metals). We sewed the wire into a cut split silicone tube (ID 0.04" OD 0.085", AM systems). The cuff was 4 mm long, with two rings of contacts, each 1 mm in length with 1 mm spacing. Each ring contained six evenly radially spaced contacts which ran parallel to the nerve. The insulation was removed to create exposed electrode contacts inside the cuff. A rendering of the cuff on the nerve is shown in figure 1.

### 2.2. Stimulator circuit

A full circuit schematic is shown in the supplement. We utilized a mirrored Howland current source as proposed by [30, 31] for each stimulator channel. We additionally included a DC blocking capacitor at the output, to prevent DC leakage during stimulation, which can skew recruitment curves [17]. We utilized MIC920 operational amplifier and 0.1% tolerance resistors to keep the output impedance high. We used DG419 switches to discharge the DC blocking capacitors after each trial. The input of each stimulator was driven by an analog output channel on a National Instruments USB 6343 Data Acquisition unit. Output currents were validated by applying the currents across known resistors and measuring the voltage.

### 2.3. Electric field models

We modeled the fields produced by various geometries using COMSOL Multiphysics 6.0. We used a steady currents model as recommended by [32]. We used constant current density boundary conditions, with each channel scaled to produce 1 mA. For techniques with unequal channel amplitudes the amplitudes were scaled such that the total current was similar to the equal-amplitude case. Each channel was modeled as completely independent and were superimposed for analysis. We used impedances based on cerebrospinal fluid for the fluid bulk, and biofidelic impedances for the other materials. Dielectric properties: conductivity ( $\text{S m}^{-1}$ ) nerve—0.0346, Pt— $9.07 \times 10^6$ , silicone— $1 \times 10^{-12}$ , fluid bulk (assumed cerebrospinal fluid)—2. Relative permittivity nerve— $4.96 \times 10^4$ , Pt—1, silicone—4, fluid



bulk—109. Values are based on 5000 Hz frequency and 37 °C [33–36]. We exported the data into MATLAB 2021b for further processing and analysis. All our model images first calculate the electric field for each independent channel. We then calculated all possible sums and differences between channels to account for possible phase difference (some locations in space are additive at 0°, while others are additive at 180°). A few images we show represent a snapshot in time, for a particular phase. Most images we show take the strongest excitation at each voxel, regardless of when in phase that excitation occurs for that voxel. Therefore, those images do not represent a snapshot in time but rather a rough estimate of firing probability over the course of the entire stimulation period.

## 2.4. Surgical procedures

All procedures were approved by the Animal Care and Use Committee of Johns Hopkins University. We present data on a total of 30 animals, with an additional 56 animals used in the development of the methods. We anesthetized both male ( $n = 16$ , weight = 364 g  $\pm$  79.6 g) and female ( $n = 14$ , weight = 255 g  $\pm$  22.1 g) Long Evans rats with urethane (1.5 g kg<sup>-1</sup> i.p.). Animals were housed in single sex pairs, with ad lib. access to food and water, under a 12 h:12 h light dark cycle. The choice of leg (left or right) was determined by a coin flip. We placed the animal prone and oriented the hip and knee at

90°. We made an incision along the skin 4 mm distal and parallel to the femur. We blunt dissected the surrounding tissue to expose the biceps femoris, which we reflected caudally to expose the sciatic nerve. We followed the sciatic nerve from the hip to the trifurcation into the tibial, sural, and peroneal nerves. We gently blunt dissected connective tissue and placed the cuff electrode just proximal to the trifurcation. We carefully tied two sutures around the cuff to keep it closed, placed at the edges of the silicone. We filled the surgical pocket with lactated Ringer's solution and covered the wound with plastic wrap to prevent it from drying out. We next expanded the subcutaneous pocket to expose the surface of the biceps femoris. We placed two 30 G stainless steel needles bent into hook electrodes, placed approximately 14 mm apart, 6 mm away and parallel to the femur, with each hook penetrating tissue approximately 2 mm wide and 2 mm deep. We placed two similar hook electrodes in the plantar of the foot, one near the ankle and one near the fifth digit. The hind limb was placed on a rigid plastic board with the animal prone, the knee and hip at 90°, the ankle at 0°, such that the foot was fully plantar flexed with the plantar facing upwards. We placed tape across the knee, ankle, and toes, just outside the recording region, in order to keep the leg in place and reduce motion artifact. We placed a stainless steel 1 inch ground plate on the skin near the base of the tail using conductive gel (Tensive).

## 2.5. Stimulation parameters

We used a custom current source stimulator (see the supplement) to drive interferential signals. Each trial consisted of 0.5 s ramp up, 1 s or 20 periods of full amplitude stimulation (whichever was longer), and 0.5 s ramp down. We randomized the order of stimulation techniques and amplitudes for a given animal. At least 20 s of rest was given between each trial. All data was processed at  $250 \text{ kS s}^{-1}$  using a NI USB-6343 DAQ and MATLAB. We used a variety of stimulus parameters which are shown in table 1.

## 2.6. Data processing

Electromyogram (EMG) data was processed using an AM-Systems 1700 (second order Butterworth; pass-band = 100–500 Hz; gain = 100; 60 Hz notch), and then digitally filtered with a 6th order Butterworth low pass filter at 650 Hz. We used MATLAB's peak detect function to find all EMG peaks during the period of maximum stimulation amplitude (1 s for most trials). For recruitment curve data the EMG amplitude of the sixth largest peak is used, to accommodate outliers. If fewer than six peaks were identified, the recruitment value was set to zero. For figures which utilize the stimulation threshold the threshold for all trials is  $25 \mu\text{V}$  of EMG activity. Neuronal maps were generated using a weighted average of 72 independent trials.

## 2.7. Statistics and error

All data presented is electromyographic muscle activity from two muscle groups. The recruitment properties of a nerve can change over the course of an experiment if the nerve is overstimulated, becomes dehydrated, or if the cuff electrodes shift relative to the nerve. Some errors are sharp and immediate (such as electrode motion), and others gradually onset (such as dehydration). To control for sharp errors all recruitment curves use a randomized stimulus order. If there is no error, the resulting curve will appear smooth. If there is an error, the curve will be jagged, and the data will be discarded (see supplement). To control for errors which gradually onset, at the end of the experiment we obtained a replicate of the first curve and compare them to ensure they are similar. The statistical comparison of the likeness of two curves can be misleading for neural data, and so we include a more thorough description in the supplement. Drift error was deemed acceptable if both the threshold did not change by more than 10% and the cumulative mean squared error between the two curves was less than the average total area under the curves. All validation curves for all experiments are shown in the supplement.

Due to practical limitations, we only tested a fixed number of stimulus amplitudes, which determines our effective resolution for determining the stimulus threshold. In data in which we plot the stimulus

threshold (figures 2–4), we show error bars. These bars represent possible error based on the sum of the uncertainty due to step size and the amount that the threshold shifted over the course of the experiment.

## 3. Results

A diagram of the cuff electrode and several types of waveforms are shown in figure 1. Animal groups are shown in table 1. For the data presented, the change in threshold over the course of the experiment was  $2.80 \pm 2.21\%$ . The uncertainty in threshold due to step size is  $3.14 \pm 1.54\%$ . The cumulative uncertainty is  $5.85 \pm 2.53\%$ . The largest possible uncertainty for any animal presented is 10.96%. In the following figures we show single examples—data from all examples is included in the supplement.

### 3.1. Effects of envelope and carrier frequencies

In four animals we explored the effect of the envelope frequency on the firing threshold (Group #1, center carrier frequency = 6000 Hz, envelope frequency  $\in [1, 1000]$  Hz). This effect has been previously modeled in [6, 10], who observed a 'U' shaped response. We have also observed this response, as shown in figure 2, albeit with a slightly different optimal frequency.

In four animals we explored the effect of the carrier frequency (Group #2, envelope frequency = 20 Hz, carrier frequency  $\in [1, 20]$  kHz) on the firing threshold. We observed a linear response. In three of these animals, we additionally determined the threshold of single sinusoids (Group #3, single sine frequency  $\in [1, 20]$  kHz). We found that the threshold was nearly identical for single sinusoids, which produce muscle tetanus and conduction block, and for interferential signals, which produce firing at the beat frequency. Data are shown in figure 2.

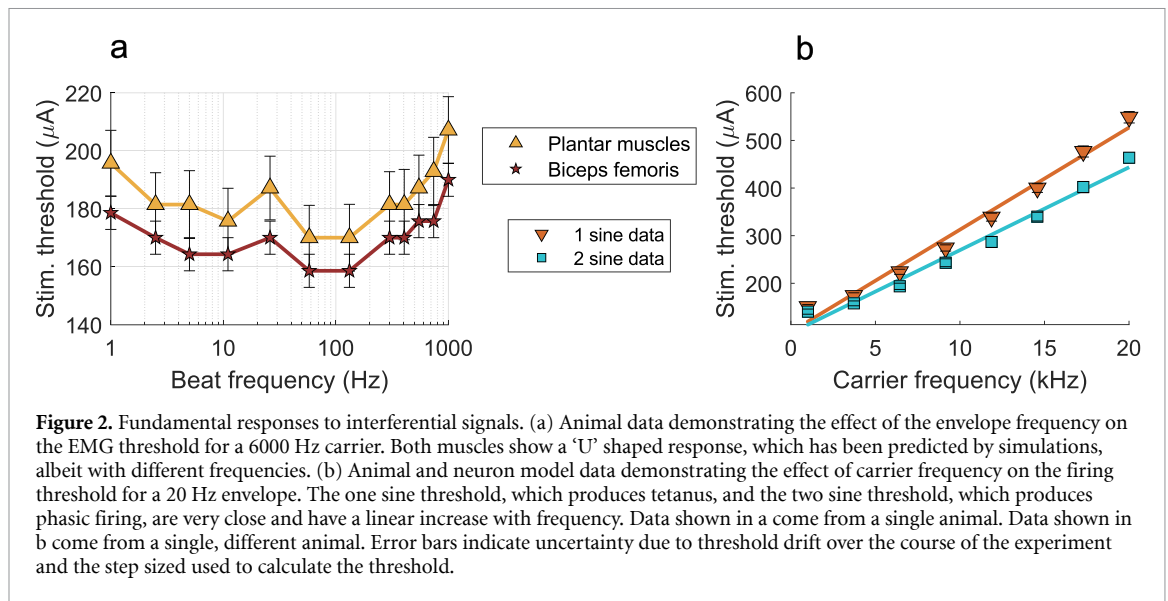
### 3.2. Undermodulation results contradict envelope extraction theory

Based on our results regarding the effects of the carrier frequency we hypothesized that neurons may not extract envelopes. To test this theory, we devised multiple strategies which maintained the same peak-to-peak amplitude for a given signal but had different envelope amplitudes (Groups #4 and #5, see figure 1, table 1). Envelope extraction predicts firing will be driven by the envelope amplitude, while an integrator theory predicts firing will be driven by the peak-to-peak amplitude. For both undermodulation (Group #4, first sine, e.g. 2000 Hz, held at a fixed amplitude, second sine, e.g. 2020 Hz, at a variable, lower amplitude) and higher order interference (Group #5, e.g. three sines 6000 Hz, 8000 Hz, 10 020 Hz, which produce 20 Hz behavior at a smaller amplitude) we observed firing thresholds consistent with an integrator, not an envelope extractor; activation

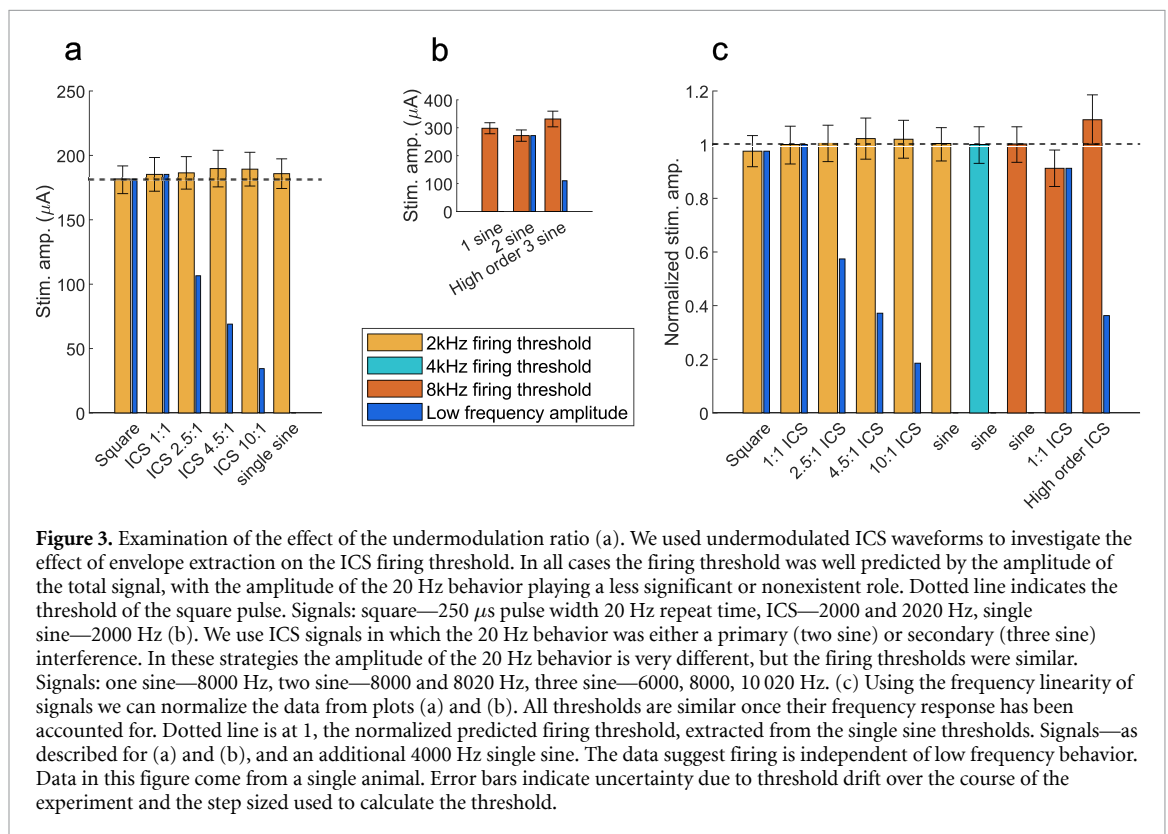


**Table 1.** Stimulus parameters.

| Group                             | No. | Signals                                                      |                                                                    |                             | <i>n</i> |
|-----------------------------------|-----|--------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------|----------|
|                                   |     | Relative amplitudes                                          | Frequency                                                          | Phase                       |          |
| Envelope frequency                | 1   | 1:1                                                          | 6000 Hz + range [1/2, 1000/2] Hz<br>6000 Hz—range [1/2, 1000/2] Hz | 0                           | 4        |
| Carrier frequency                 | 2   | 1:1                                                          | Range [1 kHz, 20 kHz] + 10 Hz<br>Range [1 kHz, 20 kHz]—10 Hz       | 0                           | 4        |
| Single sine                       | 3   | 1                                                            | Range [1 kHz, 20 kHz]                                              | 0                           | 3        |
| Undermodulation                   | 4   | 1<br>1, 2.5, 4.5, 10                                         | 2000 Hz or 8000 Hz<br>2020 Hz or 8020 Hz                           | 0                           | 3        |
| Three sine secondary interference | 5   | 1:1                                                          | 6000 Hz<br>8000 Hz<br>10 020 Hz                                    | 0                           | 3        |
| Phasic transition                 | 6   | Held at high amplitude<br>Gradually increased to match       | 2000 Hz<br>2020 Hz                                                 | 0                           | 3        |
| Phase destruction                 | 7   | Held at high amplitude<br>Gradually increased to match       | 2000 Hz<br>2000 Hz                                                 | 0<br>$\pi$                  | 3        |
| Amplitude steering                | 8   | Variable: summed amplitude preserved                         | 2000 Hz<br>2020 Hz                                                 | 0                           | 4        |
| Asymmetric envelopes              | 9   | 1                                                            | 2000 Hz                                                            | 0                           | 4        |
|                                   |     | 1                                                            | 2020 Hz                                                            | 0                           |          |
|                                   |     | 1.25                                                         | 4020 Hz                                                            | $\pi/4$                     |          |
|                                   |     | 1.67                                                         | 6020 Hz                                                            | $\pi/2$                     |          |
| Out of phase                      | 10  | Variable: summed amplitude preserved                         | 8040 Hz                                                            | $\pi$                       | 4        |
|                                   |     |                                                              | 2000 Hz                                                            | 0                           |          |
|                                   |     |                                                              | 2000 Hz                                                            | $\pi$                       |          |
| Blocking behavior                 | 11  | 1:1 (high amplitude)                                         | 2010 Hz                                                            | 0                           | 5        |
|                                   |     |                                                              | 2000 Hz                                                            | $\pi$                       |          |
|                                   |     |                                                              | 2020 Hz                                                            | 0                           |          |
| Partial block                     | 12  | Held at moderate amplitude<br>Gradually increased            | Square wave, 250 $\mu$ s PW, 10 Hz<br>1996 Hz<br>1999 Hz           | 0                           | 4        |
| Four sine secondary interference  | 13  | 1:1                                                          | 3000 Hz<br>5000 Hz<br>12 000 Hz<br>14 020 Hz                       | 0                           | 1        |
| Cao and Grover                    | 14  | 1:1 (started with 2, gradually increased to 6 total signals) | 2000 Hz<br>2005 Hz<br>2010 Hz<br>...<br>2025 Hz                    | 0                           | 3        |
| Three sine full modulation        | 15  | 1:1:1 or 1:2:1                                               | 1980 Hz<br>2000 Hz<br>2020 Hz                                      | 0                           | 3        |
| Phasic steering                   | 16  | 1:1:1                                                        | 2000 Hz<br>2000 Hz<br>2020 Hz                                      | 0<br>Range [0, $\pi$ ]<br>0 | 3        |



**Figure 2.** Fundamental responses to interferential signals. (a) Animal data demonstrating the effect of the envelope frequency on the EMG threshold for a 6000 Hz carrier. Both muscles show a ‘U’ shaped response, which has been predicted by simulations, albeit with different frequencies. (b) Animal and neuron model data demonstrating the effect of carrier frequency on the firing threshold for a 20 Hz envelope. The one sine threshold, which produces tetanus, and the two sine threshold, which produces phasic firing, are very close and have a linear increase with frequency. Data shown in a come from a single animal. Data shown in b come from a single, different animal. Error bars indicate uncertainty due to threshold drift over the course of the experiment and the step sized used to calculate the threshold.

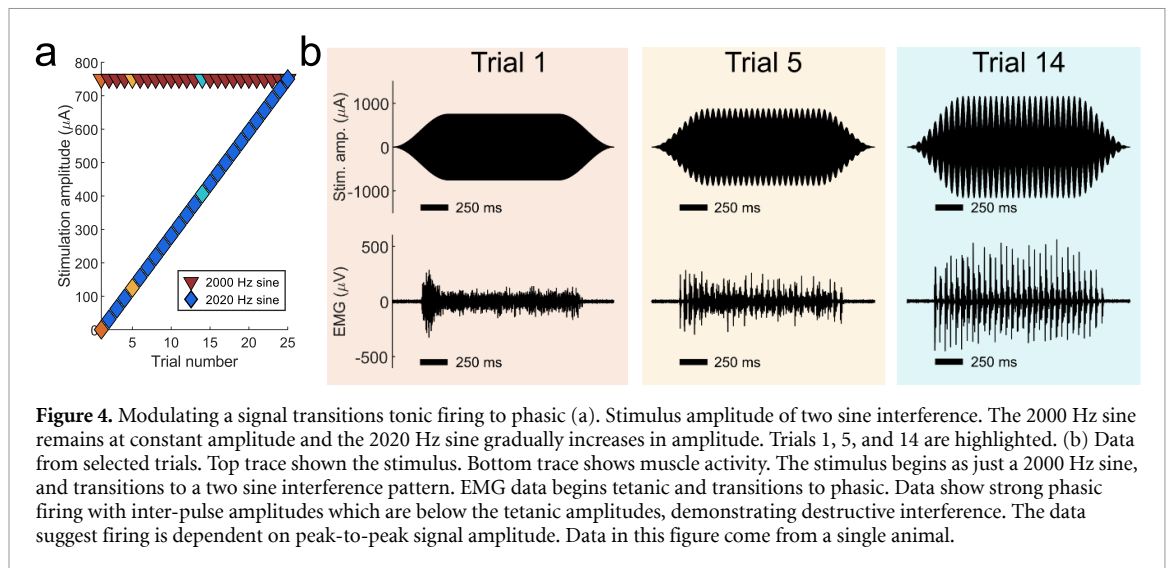


**Figure 3.** Examination of the effect of the undermodulation ratio (a). We used undermodulated ICS waveforms to investigate the effect of envelope extraction on the ICS firing threshold. In all cases the firing threshold was well predicted by the amplitude of the total signal, with the amplitude of the 20 Hz behavior playing a less significant or nonexistent role. Dotted line indicates the threshold of the square pulse. Signals: square—250  $\mu\text{s}$  pulse width 20 Hz repeat time, ICS—2000 and 2020 Hz, single sine—2000 Hz (b). We use ICS signals in which the 20 Hz behavior was either a primary (two sine) or secondary (three sine) interference. In these strategies the amplitude of the 20 Hz behavior is very different, but the firing thresholds were similar. Signals: one sine—8000 Hz, two sine—8000 and 8020 Hz, three sine—6000, 8000, 10 020 Hz. (c) Using the frequency linearity of signals we can normalize the data from plots (a) and (b). All thresholds are similar once their frequency response has been accounted for. Dotted line is at 1, the normalized predicted firing threshold, extracted from the single sine thresholds. Signals—as described for (a) and (b), and an additional 4000 Hz single sine. The data suggest firing is independent of low frequency behavior. Data in this figure come from a single animal. Error bars indicate uncertainty due to threshold drift over the course of the experiment and the step sized used to calculate the threshold.

thresholds were consistent with the peak to peak amplitude of the stimulus, regardless of the envelope amplitude. We verified these effects in three animals, and an additional two animals in which we only compared the one-sine-tetanic-activity and two-sine-phasic-activity thresholds. These data are shown in figure 3. In these experiments the signals were premodulated; the sines were summed prior to the tissue, and no interference occurred due to spatial orientation of electrodes, thus controlling for the possible spatial differences of differing amplitudes.

### 3.3. Undermodulation elimination of tonic firing contradicts envelope extraction theory

In the previous experiments we used a second signal, even undermodulated, to bring the stimulus just above the firing threshold to transition subthreshold silence to phasic firing. We can similarly introduce an undermodulated signal to bring the stimulus just below the firing threshold of a single sine to transition suprathreshold tonic firing to phasic firing. A single sinusoid at 2000 Hz of sufficient amplitude will cause tonic firing. We applied this 2000 Hz signal at a constant amplitude and gradually increased



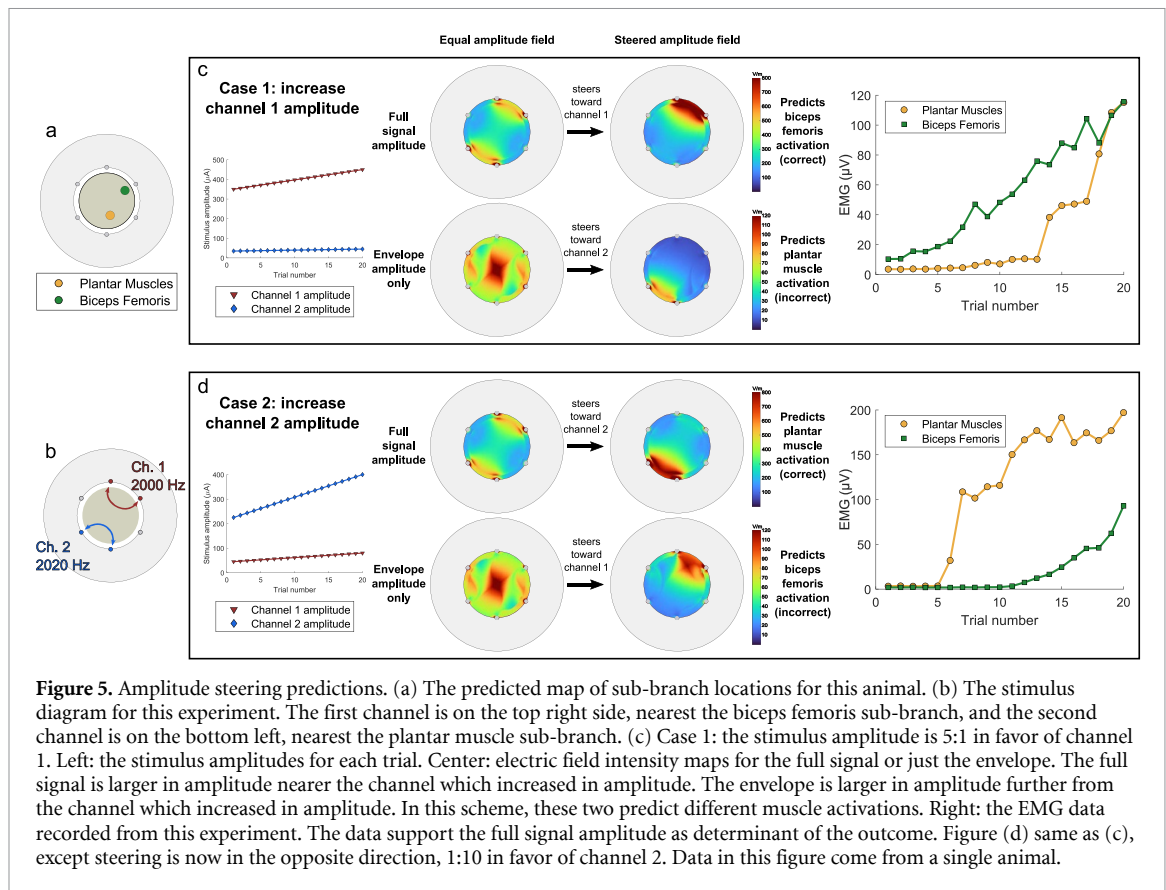
the amplitude of a second sinusoid at 2020 Hz at a spatially distinct location in the cuff (thus interference occurred inside tissue). Envelope extraction theory states that tonic firing from a single sinusoid and phasic firing from interference arise from different mechanisms at very different thresholds and would therefore predict 20 Hz phasic firing ‘on top of’ tonic firing, especially at a relatively low amplitude of the 2020 Hz signal. Instead, we observe that the tonic firing transitions to phasic firing, and the muscle tetanus is eliminated. This result contradicts envelope extraction theory and suggests that muscle firing—tonic or phasic—is the result of the integration response to a 2000 Hz signal, and by adding the second signal we are simply moving the stimulus above and below the 2000 Hz threshold, as modulated by the 2020 Hz signal, analogous to pulsing the 2000 Hz stimulus at 20 Hz. We verified this result in three animals (Group #6). These data are shown in figure 4. In another three animals we applied a 2000 Hz sine at sufficient amplitude to cause tetanic firing, and then gradually increased the amplitude of another 2000 Hz sine 180° out of phase at a spatially distinct location in the cuff (Group #7). Unsurprisingly, the two signals interfered in space inside the nerve, and tetanic activity was eliminated, without producing phasic activity.

### 3.4. Amplitude steering results contradict envelope extraction theory

In another set of experiments, we explored the claims of spatial steering with varying amplitudes. One benefit of ICS is that the stimulation field is steerable—the targeted neuronal region can be moved because the region is created by multiple signals originating from different points in space. The most proposed method of steering is changing the relative amplitudes of the signals. For example, if we are using a 2000 Hz signal on the left side of a nerve and a 2020 Hz signal

on the right side of the nerve, then activation should be symmetric about the center of a nerve. If neurons extract envelope energy in a significant amount then if we increase the amplitude of the 2020 Hz signal the region of activation will move towards the 2000 Hz signal, since activation is dependent on both signals being present. However, if only the total signal amplitude matters then an increase in the amplitude of the 2020 Hz signal will move the region of maximal activation toward the 2020 Hz signal. Finally, if both types of signals are equally important, we would expect little change in recruitment. In four animals we first obtained multiple measurements that allowed us to estimate the location of each motor group within the nerve. Next, we used two sines to cause activation, and increased the amplitude of one or the other sine in order to steer the signal (Group #8, two sines, 2000 Hz, 2020 Hz, with variable amplitudes to cause spatial steering). In all four animals steering causes an increase in motor activity in the motor group that was closer to the signal that was increased. These data are shown in figure 5. These results are consistent with the total signal amplitude predicting outcomes, and envelope energy extraction playing a minor role. In 1/4 animal we continued to change the relative ratios of these two sines ranging from a 1:1 ratio to an infinite ratio, stimulating with only a single sine. When the signal transitions from two sines to one sine the envelope ‘disappears.’ If neurons significantly extract envelope energy, we would expect to see a sharp difference in motor activity at this transition. If neurons do not extract significant envelope energy we would expect little to no change in activity at this transition. We observed no significant change in threshold when moving from a strong steer to a single sine. We also observed motor activity transitions from phasic to tonic, suggesting motor activation is primarily driven by the high amplitude single sine, not envelope energy (see supplement).





### 3.5. Asymmetric envelopes

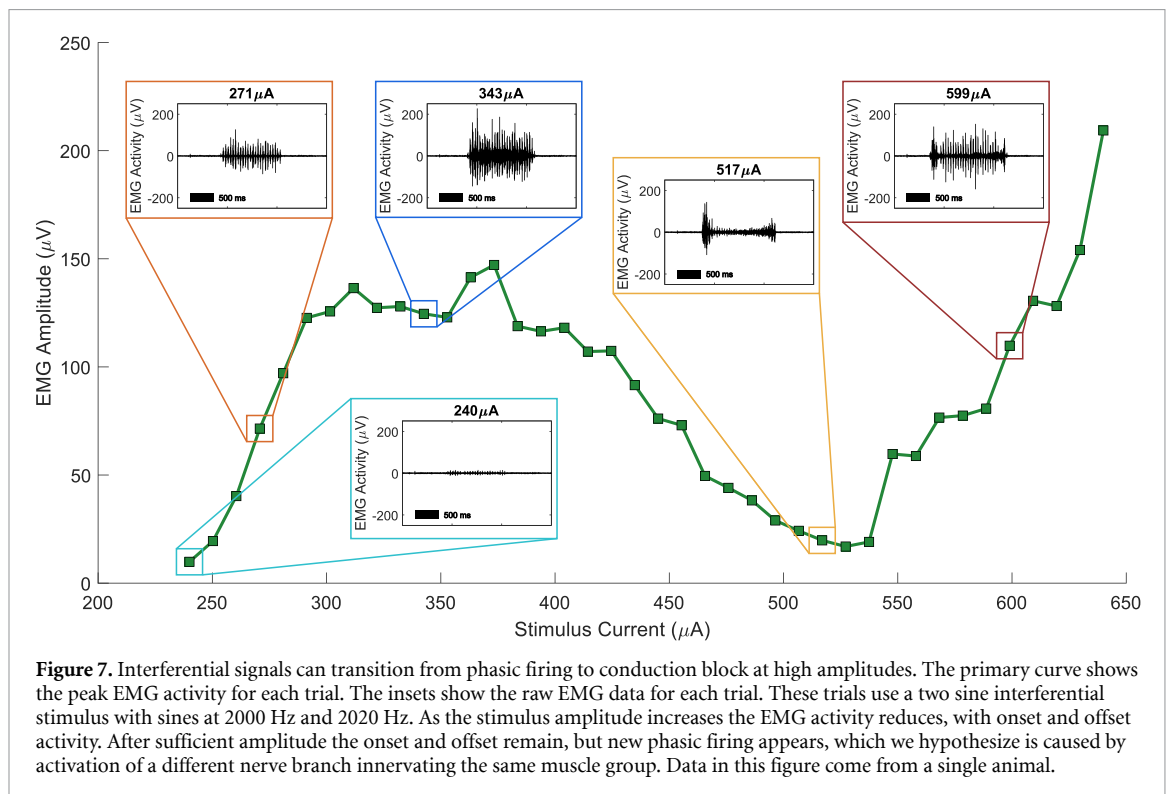
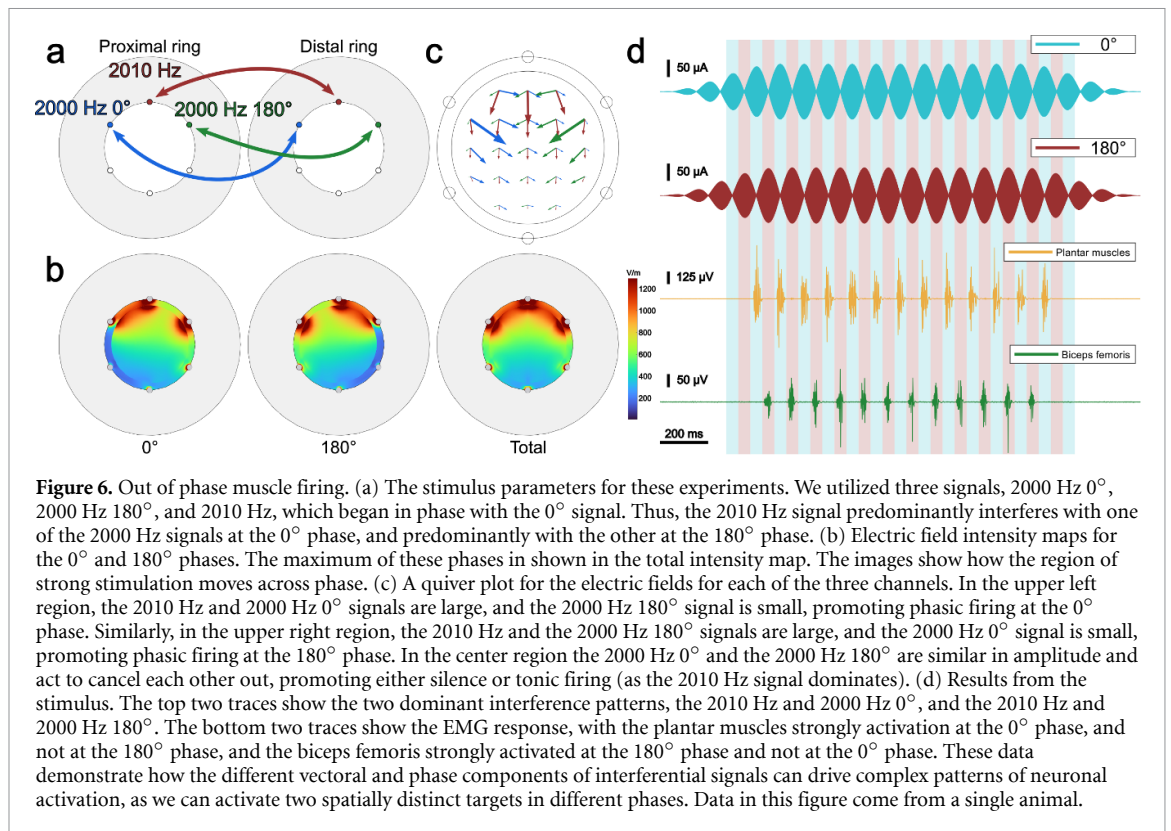
We explored another open question regarding the mechanism of interferential stimulation: does the nerve respond preferentially to the top envelope, bottom envelope, or both? The envelope at the top and bottom of a signal need not be identical in amplitude, frequency, or phase. Further, the sign of a voltage waveform flips over the equipotential line between two electrodes. We know that the magnitude of the change in ion channel activation probability is larger for depolarizing signals than it is for hyperpolarizing signals of the same amplitude, which is the basis of cathodal first stimulation parameters. Given our data supporting an integrator mechanism we hypothesized this property would extend to interferential signals, and that nerves will preferentially respond to the cathodal bottom (i.e. cathodal depolarizing) envelope. We devised a charge-balanced waveform that had very different depolarizing and hyperpolarizing peaks (see figure 1). We then applied this signal and its inverse and observed significant differences in motor recruitment in four animals (Group #9, see table 1 for full signal list), with firing more prominent in the region of cathodal stimulation. When we model this effect, we observe a sharp discontinuity because we are only interested in the most depolarized potential for each location in space, which flips along the equipotential line at zero voltage. These data are shown in figure 9.

### 3.6. Out of phase firing driven by differing components

Terasawa *et al* demonstrated that two interfering sinusoids produce amplitude modulated shapes in the different electric field vector components that occur out of phase with each other [13]. That is to say, it is possible to design a stimulus such that certain regions of interference are depolarizing at the  $0^\circ$  phase, while other regions are depolarizing at the  $180^\circ$  phase. We hypothesized that this property could be utilized to cause different regions of the nerve to fire out of phase with each other. We further emphasized this effect by using three sinusoidal signals: two at 2000 Hz with a  $180^\circ$  phase difference, and the third signal at 2010 Hz (Group #10). By carefully choosing the geometry and relative amplitudes of signals, we were able to achieve  $180^\circ$  out of phase firing in the two target muscles in four animals (see figure 6). By driving both sinusoids in phase or by changing which contacts current was driven from, the muscles could be activated in phase as well. Therefore, interferential signals can activate geometrically distinct targets according to different phases. Presumably, this ability extends to different frequencies as well.

### 3.7. Blocking behavior

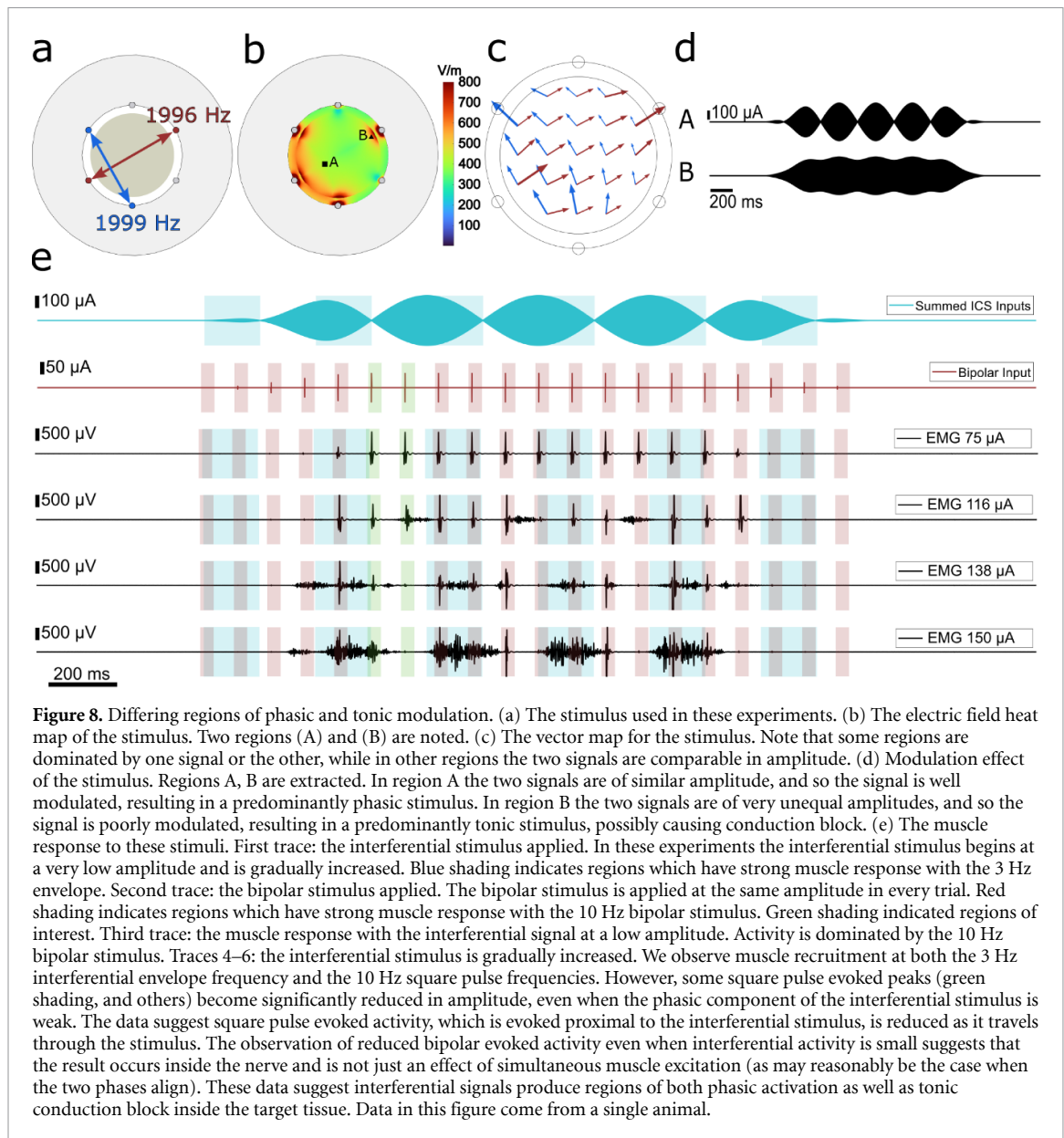
Mirzakhali *et al* predicted that higher amplitude ICS signals can cause conduction block, especially when the component signals are applied from different



locations in space [10]. In five animals we increased the stimulation amplitude of ICS signals well beyond what was necessary to elicit motor contractions (Group #11). As stimulation amplitude increased, we observed a gradual decrease in motor activity until we only observed onset and offset firing, consistent with conduction block. In two animals we also saw

a subsequent increase in activity, as we hypothesize that we started to activate an adjacent neuronal track which innervated the same muscle. These data are shown in figure 7.

In four animals, we placed a bipolar cuff electrode proximal to the 12-contact interferential cuff electrode. We used this bipolar cuff electrode

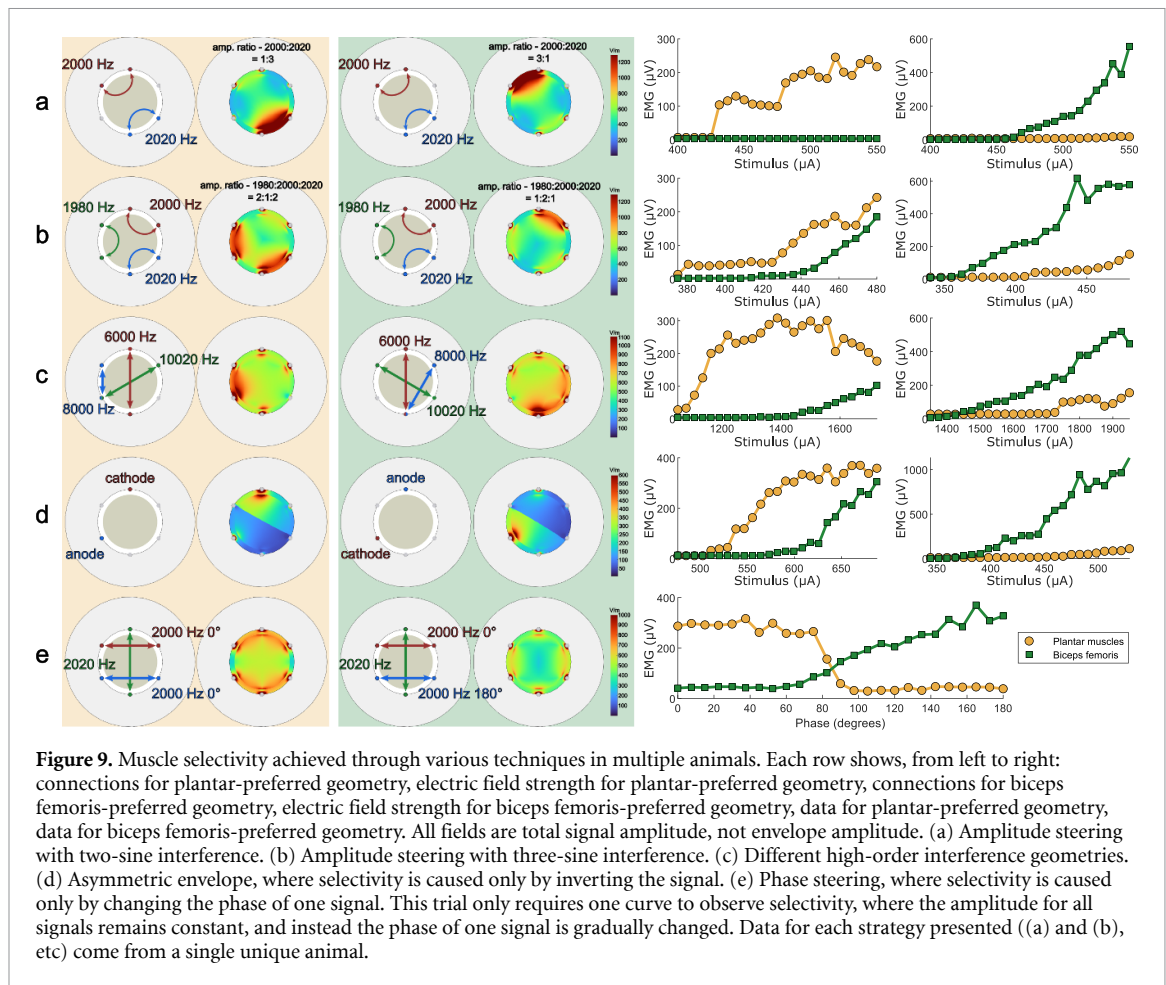


to stimulate the nerve with traditional biphasic square pulses proximally while concurrently stimulating with ICS waveforms distally using the 12-contact electrode (Group #12; bipolar—250  $\mu$ s pulse width, 10 Hz repeat time; ICS—1996 Hz and 1999 Hz). We used different beat frequencies at our interferential signals (3 Hz) compared to the frequency of bipolar stimulation (10 Hz), so muscle contractions at different times could be associated with the two different stimuli. We first determined the bipolar stimulation threshold and applied that stimulus at a constant amplitude. Next, we gradually increased the amplitude of the interferential signal. At lower amplitudes of the interferential signals, we observed muscular activity at both frequencies, with the bipolar-evoked activity remaining similar to baseline. However, as the amplitude of the interferential signals was increased, the amplitude of the

bipolar-evoked activity became reduced, even in phases when the interferential signals were destructively interfering and not producing phasic muscle firing, suggesting that the interferential signals may have had a blocking effect in parts of the nerve (see figure 8).

### 3.8. Firing thresholds are consistent with peak amplitude regardless of interference type

In four animals we explored the firing behavior in response to signals with ‘secondary interference’ (Groups #5 and #13), at which the low frequency envelope is not the primary envelope behavior. Consider Group #5 with signals at 6000 Hz, 8000 Hz, and 10 020 Hz. The first two signals interfere at 2000 Hz, the second two signals interfere at 2020 Hz, and the three therefore have a small interference at 20 Hz. Similarly, Group #13 uses signals 3000 Hz, 5000 Hz,

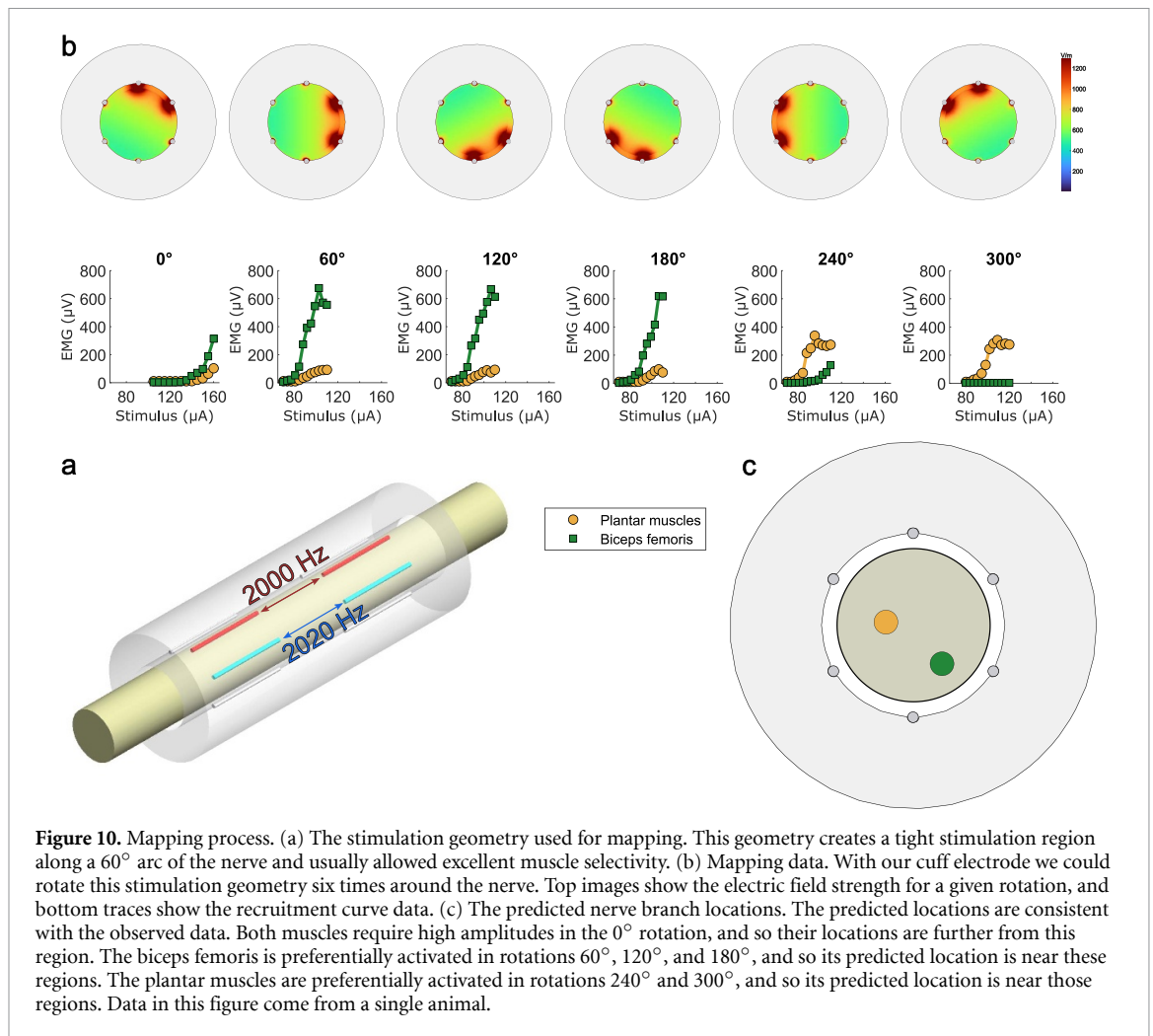


12 000 Hz, 14 020 Hz. We observed muscle activation at the beat frequency of 20 Hz for both groups. In these signals, the 20 Hz envelope amplitude is much smaller than the peak-to-peak amplitude. The firing thresholds for these techniques were consistent with the peak-to-peak amplitude, not the 20 Hz behavior, supportive of an integration mechanism.

In three animals we utilized a technique proposed by Cao and Grover [6] (Group #14), which uses six sinusoids at 2000 Hz, 2005 Hz, 2010 Hz, etc up to 2025 Hz. We started with just two sinusoids at 2000 Hz and 2005 Hz, and then added a third at 2010 Hz, and then 2015 Hz, etc. We observed that in each case the firing threshold is consistent with the total current input (see supplement). We predict this strategy will be useful for optimal interferential strategies as it maximizes the interference amplitude while minimizing the amplitude of each contributing signal. Similarly, in three animals we attempted three-sine interference (Group #15; 1980 Hz, 2000 Hz, 2020 Hz) which produces full interference of a pure sinusoid, unlike the more common two-sine interference which produces interference of a rectified sine wave (figure 1). We observed no significant improvements with the addition of the third sine, and its threshold remained consistent with the total peak to peak amplitude input.

### 3.9. Geometric targeting and mapping

Many of the techniques and phenomena we have discussed so far might be used to cause a spatial change in the recruitment of a target. In a series of experiments we utilized these techniques to target the different muscles, demonstrating steering techniques beyond amplitude steering. We used two sine interference ( $n = 2$ ), three sine interference ( $n = 3$ ), higher ordered interference ( $n = 4$ ), and asymmetric envelopes ( $n = 4$ ), along with complex geometries to change muscle recruitment. We used phase interference ( $n = 3$ ) without changing the geometry to similarly target muscle recruitment (Group #16). Because the signals are all generated from different contacts there are multiple regions of constructive and destructive interference, allowing the generation of both stimulation ‘hot spots’ and ‘cold spots.’ Singular examples are shown in figure 9. Additional examples are shown in the supplement. In order to aid in the targeting of these muscle groups, we developed a simple mapping procedure to estimate the locations of the two major branches of the nerve. We utilized one geometry, our mapping geometry, which targeted a  $60^\circ$  slice of the nerve near the edge of the cuff. By rotating this geometry around the cuff across multiple trials we were often able to obtain good separation between our two muscle targets. Using these



data, we generated an estimated map of each target within the nerve, and then utilized this map to inform subsequent trials and optimize targeting. An example of the mapping process is shown in figure 10. We collected well separated maps in seven animals, shown in the supplement.

#### 4. Discussion

In this experimental setup we explore the relative contribution of nonlinear, envelope extracting membrane dynamics in peripheral nerve stimulation with interferential stimuli. Across multiple experimental types we found that nonlinear, envelope extracting membrane properties contribute little, if at all, to the firing behavior. Instead, we found that the neuronal response to interferential signals is well predicted by a linear integrate and fire type model. These results suggest that interferential stimuli, contrary to previous claims, cannot modulate a central target, far from electrodes, without first modulating periphery targets, near electrodes. Therefore, our data suggest that interferential techniques may not be as capable at noninvasive deep brain stimulation as previously proposed. However, our data also suggest

that while the locations of possible modulation are similar to traditional techniques, interferential stimuli can allow the type of modulation to vary across the stimulus region. We were able to produce spatially distinct regions of tetanic activity, conduction block, phasic activity at the beat frequency, double the beat frequency, and at varying phases. Therefore, while it is not possible to only modulate a central target, it may be possible to modulate a central target differently than other targets. We used a peripheral, large diameter, myelinated axon where firing is primarily driven by voltage gated sodium channels at nodes of Ranvier. We have not conclusively validated these conclusions in other scenarios, namely voltage gated potassium and calcium channels, or the synapses, dendrites, soma, or unmyelinated axons. However, given the known biophysics of these various settings, and their similarity to our experiments, we believe conclusions will be similar. We do not prove or disprove if envelope extraction occurs, but our data show that if it does occur, it contributes to firing behaviors minimally (approximately less than our 6% threshold uncertainty).

Utilizing an integrator-threshold mechanism, the neuromodulation capabilities of TIS are reduced



compared to previous claims, but they may still greatly improve over non-interferential techniques. We believe that interferential techniques are still worthy of investigation and may improve a broad range of neuromodulation techniques:

1. Interferential techniques allow moderate control over the volume of the region of neuronal modulation. This region is the superposition of single sources, similar to current steering currently capable with non-interferential techniques [37].
2. Interferential techniques allow moderate control over the frequency and phase of modulation. Different regions of tissue can be activated according to different envelopes, and therefore fire according to different frequencies and phases.
3. Interferential techniques allow significant control over the type of modulation. Traditional bipolar techniques will only stimulate the entire region at the desired frequency. Because interferential techniques use high frequency carriers (over 1 kHz) each carrier produces tonic firing or conduction block on its own, but phasic firing when combined with other signals. Therefore, interferential stimuli can produce spatially distinct regions of phasic firing, conduction block, tonic firing, and no effect. This property includes the ability to block neurons near electrodes and cause phasic firing in neurons far from electrodes. In some scenarios this ability is nearly as effective as TIS as originally promised.

Upon initial inspection, our results of amplitude steering appear to contradict the results demonstrated by Grossman *et al* [9]. In our results increasing the amplitude of a single channel increased phasic activity near that channel. In the other study the same technique increased phasic activity near the other channel. Indeed, both outcomes are possible depending on the amplitudes used relative to the stimulation threshold. As we demonstrate in figure 8, it is possible to produce regions of phasic activity, which is easily observed and measured, as well as regions of tonic activity/conduction block, which is more difficult to measure and can be missed if not explicitly looked for, especially using video analysis. Our steering data is relatively conclusive because we measure EMG activity and because we begin at amplitudes that are subthreshold for all measurable activity and then gradually increase. The first observation is an increase in phasic firing in the target that is near to the channel that was increased. It is possible that if we had increased our stimulation to higher amplitudes we may have produced tonic firing/conduction block in the target near the channel that was increased, which may be undetectable on video. In a high-amplitude case we may also see phasic firing in the target

further from the channel that was increased, giving the appearance of steering moving in the opposite direction, giving the appearance of envelope extraction. Further, our data is consistent when steering in the opposite direction with similar amplitudes.

Our data imply several fundamental properties of voltage gated channels and neuronal excitability. The data suggest individual ion channels do not extract envelopes. Rather, behavior at a beat frequency is a consequence of aggregate behavior from a population of channels. The generation of the action potential then acts as a nonlinear operator which produces beat behavior. Therefore, spatial superposition of signals must occur before ion channel dynamics are considered. The data suggest the membrane acts as a near-ideal integrator, even at high frequencies (tested up to 20 kHz), and that ion channels can respond to these fast dynamics in a linear fashion. Therefore, interferential techniques may allow manipulation of ion channel behaviors without impacting the whole cell. These mechanisms imply interferential techniques work by taking advantage of the onset firing observed with high frequency stimuli, and modulating the stimulus such that onset firing behaviors are pulsed, giving the appearance of a lower frequency stimulus.

Our data includes several limitations. Due to the practical experiment duration, we were only able to test a limited number of parameters on each nerve, and therefore the data includes some uncertainty regarding the exact threshold for each nerve (approximately 6%). In order to investigate the spatial properties of these techniques we used muscle activity as a proxy for neural activation, which is imperfect as muscle activity is subject to behaviors not found in neurons, such as fatigue. In most cases our sample sizes are small for each protocol of investigation, but this concern is partially ameliorated because some protocols support others. For example, the threshold observations of undermodulation, asymmetric envelope, single sines, four sine secondary interference, and near-frequency interference all support one another, and investigate a similar underlying mechanism. Our goal in this study was merely to demonstrate the existence of certain phenomena and general trends of recruitment properties, as these are sufficient to draw conclusions about the underlying mechanisms of these stimuli. These data are insufficient to make strong claims about specific quantitative properties. For example, we show data that suggests that, with regards to the carrier frequency, the two-sine-interferential threshold and the single-sine threshold are approximately similar and approximately linear, but we do not have adequate data to determine the degree of similarity, or what the slope of the linearity should be.

Interferential techniques may still improve the state of the art for neuromodulation. Interference

techniques allow for novel stimulation patterns not possible with traditional bipolar neurostimulation such as localized conduction block in conjunction with phasic stimulation. Such a stimulation pattern could be useful in treating conditions such as urinary incontinence or spinal cord injury where both localized stimulation and inhibition are desired [38]. So far we have only tested each parameter of interference in isolation. Interferential neurostimulation is a promising technology because the complexity of these techniques is multiplicative. We can combine complex geometry via the 12 contacts and phase induced hot and cold spots and higher order interference patterns and asymmetrical interference. These techniques are also only targeting neurons by their physical location. By optimizing the envelope frequency, we might also target neurons based on their diameter, a technique already used in pulse-width optimization of fiber type recruitment [39]. Our work exclusively used sinusoids for simplicity, but the integration and interferential properties should extend to waveforms of arbitrary shape. Lastly, interferential techniques can be used with any type of wave, including optical, sound, or magnetic. Investigation into interference in these domains may also yield a significant improvement in neuromodulation practice.

### Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://doi.org/10.7281/T1/VZYIRV> [40].

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### Author contributions

R B B and M T W conceived the study, designed and performed all experiments, conceived and designed all tools, methods, code, and techniques, analyzed data, generated figures, and wrote the manuscript. P P I contributed to study conception, experimental design, and reviewed the manuscript.

### Conflict of interest

The authors have applied for a patent for the cuff electrode design: 63/280 652, INTERFERENTIAL CURRENT STIMULATOR FOR PERIPHERAL NERVES.

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